

FINANCIAL CRIME AND FRAUD INVESTIGATION POLICY

AI-Assisted Transaction Monitoring, Alert Triage & Case Disposition Framework

Document ID:	COMP-FRD-2026-V2
Version:	2.0 (Supersedes V1.0)
Classification:	Confidential – Internal Use Only
Effective Date:	July 15, 2026
Review Cycle:	Annual (or upon material regulatory change)
Prepared by:	Fraud Prevention & AML Compliance Department
Approved by:	Chief Compliance Officer & Board Risk Committee

This policy establishes mandatory protocols for AI agents and human analysts engaged in the detection, investigation, escalation, and final disposition of potentially fraudulent, suspicious, or high-risk financial transactions. It applies to all payment rails processed by Nexus Financial Technologies platforms, including card-present / card-not-present transactions, wire transfers, ACH, instant payments, and digital wallet activity.

IMPORTANT: All AI-generated case files must strictly adhere to the 3-bullet summary + final recommendation format defined in Section 4.4. Non-compliance constitutes a policy violation subject to model governance review.

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1. PURPOSE AND OBJECTIVES

The primary objective of this policy is to provide a robust, auditable, and regulatorily compliant framework for the hybrid investigation of financial crime and fraud alerts. By clearly defining the interaction between automated machine learning models (Isolation Forest for anomaly detection, XGBoost for probability scoring, and PCA-based feature engineering) and human oversight, the Company achieves the following strategic outcomes:

- Minimize financial losses from fraudulent transactions while maintaining a low false-positive rate that does not unduly impact legitimate customer activity or create excessive operational burden.
- Ensure consistent, timely, and well-documented disposition of alerts in accordance with internal risk appetite and external regulatory expectations (including FATF Recommendations, BSA/AML requirements, and applicable local financial crime legislation).
- Reduce alert fatigue among human analysts by empowering AI agents to auto-dismiss low-risk, low-value outliers that meet strict quantitative criteria.
- Guarantee that every high-risk or high-value case receives appropriate human review and, where warranted, escalation to the specialized AML Compliance Team for Suspicious Activity Report (SAR) consideration or law enforcement referral.
- Maintain full auditability of all AI decisions, human overrides, and final recommendations to support internal model risk management, external audits, and regulatory examinations.

2. SCOPE AND APPLICABILITY

This policy applies to all automated alerts generated by the Company's real-time and batch transaction monitoring systems. It governs the behavior of AI Investigation Agents, human Fraud Analysts (Tier 1 and Senior), the AML Compliance Team, Model Risk Management, and any third-party vendors performing outsourced alert investigation functions.

Transaction Types Covered: Credit and debit card transactions (CP/CNP), domestic and international wire transfers, ACH credits/debits, instant payment schemes (e.g., FedNow, RTP), digital wallet top-ups and P2P transfers, and any other payment rail processed through the Company's acquiring or issuing platforms.

Geographic Scope: All transactions involving customers or counterparties domiciled in the United States, its territories, and any cross-border transactions where the Company acts as the originating, intermediary, or beneficiary financial institution.

3. DEFINITIONS AND KEY TERMINOLOGY

The following definitions apply throughout this policy:

PCA Features (V-14, V-10)

Principal Component Analysis (PCA) transformed variables derived from the raw transaction feature set. V-14 typically captures variance associated with transaction velocity, amount deviation from customer baseline, and merchant risk clustering. V-10 often loads on temporal and geographic anomalies. These components are used as high-signal inputs to both Isolation Forest and XGBoost models.

Tier 1 High-Risk Anomaly

Any transaction exceeding USD 5,000 that triggers an alert on PCA feature V-14 or V-10 (or both). Such cases receive the highest priority workflow and mandatory senior analyst review.

Isolation Forest

Unsupervised machine learning algorithm used to detect outliers in high-dimensional transaction data without requiring labeled fraud examples. It isolates anomalies by randomly partitioning data; shorter path lengths indicate higher anomaly scores.

XGBoost Probability Score

Output (0–100%) from the gradient-boosted supervised classifier indicating the estimated probability that a given transaction is fraudulent. Scores are calibrated on historical labeled data and refreshed periodically.

Daily Average Spending

Customer-specific baseline calculated as the median transaction amount over the preceding 90 calendar days (rolling window, excluding known fraud and verified one-off high-value purchases). The median is preferred over the mean to reduce sensitivity to outliers.

Velocity Alert

Flag generated when one or more transactions in a short window cause the customer's current spending rate to exceed 300% of their established daily average baseline.

Travel Notice

Customer-initiated, time-bound notification (via app, web portal, or customer service) declaring upcoming travel dates, destinations, and expected transaction patterns. Valid travel notices reduce the risk score of otherwise anomalous night-time or foreign transactions.

AML Compliance Team

Specialized second-line unit responsible for conducting enhanced investigation of SAR-narrative cases, filing timely Suspicious Activity Reports with FinCEN (or equivalent regulator), coordinating with law enforcement, and maintaining the Company's financial crime risk appetite framework.

Auto-Dismiss

Final disposition status assigned by an AI Agent when all strict criteria in Section 4.3 are met. The case is closed without human review but remains subject to retrospective quality sampling.

Hold Status

Immediate system action that prevents a transaction from being authorized, cleared, or settled pending completion of the mandated investigation workflow.

4. POLICY REQUIREMENTS AND OPERATIONAL PROCEDURES

4.1 High-Value Anomalies (V-14 & V-10 Triggers)

4.1.1 Detection and Immediate Action

Any transaction with a USD (or USD-equivalent) amount exceeding \$5,000 that generates an alert on PCA feature V-14 or V-10 must be automatically placed in **HOLD** status by the transaction authorization engine. The investigating AI Agent must classify the alert as a “**Tier 1 High-Risk Anomaly**” within its initial case file.

4.1.2 Dual-Trigger Escalation (V-14 AND V-10)

When both V-14 and V-10 features are triggered simultaneously on the same transaction, the AI Agent **must** include the explicit recommendation “**ESCALATE TO AML COMPLIANCE TEAM**” in the final case report. This dual-trigger scenario is treated as presumptively indicative of sophisticated layering, structuring, or account takeover involving high-value movement.

4.1.3 Investigation Workflow and SLAs

- AI Agent performs automated enrichment (customer 12-month transaction profile, device fingerprint trust score, geo-IP vs. registered address, merchant category risk, prior alert history, PEP/adverse media flags) and generates preliminary case file within 5 minutes of alert creation.
- Case is routed to the on-duty Senior Fraud Analyst queue with a 30-minute SLA during core business hours (08:00–20:00 ET) and a 2-hour SLA during off-hours.
- Senior Analyst must document decision rationale, any customer outreach performed, and final disposition. All Tier 1 cases require secondary review by a Team Lead before closure or escalation.
- If the analyst determines the transaction is legitimate (e.g., verified large purchase with supporting travel notice or known vendor payment), they may release the hold and close the case as “FALSE POSITIVE – VERIFIED LEGITIMATE”.
- If red flags persist (inconsistent customer explanation, rapid movement to high-risk jurisdiction, source of funds unclear), the case must be escalated to the AML Compliance Team with a complete handoff package.

Example: A \$12,400 wire to a newly onboarded beneficiary in a high-risk jurisdiction at 02:17 local time triggers both V-14 (velocity deviation) and V-10 (temporal/geographic anomaly). The AI Agent must immediately recommend escalation to AML and populate enhanced due-diligence fields (source of wealth questionnaire link, adverse media summary, and prior SAR history check).

4.2 Velocity and Time-Based Fraud Detection

4.2.1 Flagging Logic

Transactions occurring between **00:00 and 04:00** (account holder's local time) whose amount exceeds the customer's established daily average spending by **300% or more** (i.e., $\geq 4\times$ baseline) must be flagged for mandatory manual review. The AI Agent must first retrieve and evaluate any active Travel Notice on file.

4.2.2 Contextual Enrichment Requirements

Before recommending a disposition, the AI Agent shall automatically collect and surface the following contextual data points:

- Existence and validity period of any Travel Notice; cross-reference with transaction geo-location and merchant country.
- Device fingerprint trust score and any recent device enrollment or password/PIN reset events (within preceding 7 days).
- Merchant category code (MCC) risk tier – gambling, cryptocurrency exchanges, money services businesses, and high-risk e-commerce categories receive elevated scrutiny.
- Velocity within the preceding 2-hour and 24-hour windows compared with the customer's historical pattern.
- Whether the transaction is the first international or high-value transaction for this customer in the preceding 90 days.

4.2.3 Disposition Guidance

If a valid, matching Travel Notice is on file and geo-location is consistent, the AI Agent may recommend "MONITOR – TRAVEL NOTICE VERIFIED" with a lower priority queue placement. All other cases in this category require human analyst review within the applicable SLA. Night-time high-velocity activity without travel notice is a classic indicator of account takeover (ATO) or authorized push payment (APP) fraud and must not be auto-dismissed under any circumstances.

4.3 False Positive Dismissal Criteria

To combat alert fatigue and optimize human analyst capacity, the following **strict conjunctive criteria** permit an AI Agent to assign an "AUTO-DISMISS" recommendation without human review:

- The alert was generated solely by the Isolation Forest model (no concurrent XGBoost score above 40% and no V-14/V-10 Tier 1 trigger).
- The XGBoost fraud probability score is **strictly below 40%**.
- The transaction amount is **less than USD 50** (or local currency equivalent).
- No additional high-risk indicators are present (new payee added within 24 hours, first-time cross-border transaction, high-risk MCC, customer on enhanced monitoring list, or recent dispute history).

Quality Assurance Override: The Fraud Operations Team Lead shall conduct a weekly random sample of 5% of all AUTO-DISMISS cases. Any systemic pattern of missed fraud in the low-value band must trigger immediate model recalibration and policy review.

4.4 Mandatory Reporting Format for AI Agents

4.4.1 Standardized Case File Conclusion

Every automated case file generated by an AI Investigation Agent **must** conclude with exactly three bullet points followed by one of the three permitted final recommendations. No free-text narrative, additional analysis, or hedging language is permitted after the structured summary. All supporting enrichment data and model scores must appear in the “AI Enrichment Notes” section that precedes the mandatory summary.

Required Structure (exact order and formatting):

CASE SUMMARY

- **Trigger & Quantitative Profile:** Transaction amount \$X,XXX.XX; Isolation Forest anomaly score: XX; XGBoost probability: XX%; PCA V-14 contribution: XX; PCA V-10 contribution: XX; Local time: HH:MM; Velocity vs. 90-day baseline: +XXX%.
- **Contextual Factors:** [Travel notice status / Device trust score / Merchant risk tier / Geo consistency / Recent account changes / Prior alert history – one or two sentences maximum].
- **Initial Risk Assessment:** [Concise statement of why the transaction raises or does not raise concern, including any immediate system actions such as HOLD placed].

FINAL RECOMMENDATION: [Exactly one of: ESCALATE TO HUMAN | ESCALATE TO AML COMPLIANCE TEAM | AUTO-DISMISS]

4.4.2 Permitted Recommendations

Recommendation	When to Use	Next Action
ESCALATE TO HUMAN	Tier 1 High-Risk Anomaly, velocity/time-based flags, or any case not meeting auto-dismiss criteria	Route to Senior Fraud Analyst queue with applicable SLA
ESCALATE TO AML COMPLIANCE TEAM	Dual V-14 + V-10 trigger, suspected structuring, high-risk jurisdiction movement, or other indicators meeting SAR threshold	Immediate handoff to AML queue; AI populates enhanced SAR fields
AUTO-DISMISS	Strictly meets all four criteria in Section 4.3; low-value, low-confidence Isolation Forest outlier only	Case closed automatically; 5% weekly QA sample by Team Lead

5. ROLES, RESPONSIBILITIES AND ESCALATION MATRIX

5.1 Primary Roles

AI Investigation Agents – Execute real-time policy logic, perform data enrichment, generate standardized case files, and apply AUTO-DISMISS where all quantitative criteria are satisfied. AI Agents operate under model governance oversight and may not deviate from the four policy sections above.

Human Fraud Analysts (Tier 1) – Review all cases escalated by AI Agents (except pure AUTO-DISMISS). Conduct customer outreach when necessary, document rationale, and either close the case or escalate to Senior Analyst / AML Team. Must complete mandatory annual financial crime training and pass scenario-based assessments.

Senior Fraud Analysts / Team Leads – Own Tier 1 High-Risk Anomaly cases, perform secondary review of complex dispositions, conduct the 5% AUTO-DISMISS quality sample, and approve policy exception requests within their delegated authority.

AML Compliance Team – Receive escalated cases meeting SAR criteria, perform enhanced investigation (source of funds/wealth, network analysis, adverse media), prepare and file SARs/CTRs within regulatory deadlines, and maintain liaison with FinCEN, FBI, and local law enforcement.

5.2 Escalation Matrix (Summary)

Trigger Category	Initial Owner	SLA	Escalation Path
Tier 1 High-Value (\$5k+ V-14/V-10)	AI → Senior Analyst	30 min (business) / 2 hr (off)	AML Team if dual trigger or SAR indicators
Night Velocity ≥300% above baseline	AI → Human Analyst	1 hour	Senior Analyst if ATO suspected
Low-value Isolation Forest only (<\$50, XGBoost <40%)	AI Agent (Auto)	Immediate	5% weekly QA by Team Lead
Any case with SAR-narrative indicators	Human Analyst	4 hours	Direct to AML Compliance Team

6. GOVERNANCE, AUDIT AND CONTINUOUS IMPROVEMENT

6.1 Policy Ownership and Review

This policy is owned by the Head of Fraud Prevention, with joint accountability to the Chief Compliance Officer. It shall be reviewed at minimum annually, or more frequently following any material fraud incident, regulatory enforcement action, or significant model performance drift (defined as >15% degradation in precision or recall on the hold-out test set).

6.2 Model Risk Management Integration

All changes to Isolation Forest contamination rate, XGBoost decision threshold, PCA feature weights, or the quantitative criteria in Sections 4.1–4.3 require formal Model Risk Management approval and documentation in the model inventory. Back-testing of policy effectiveness (true positive rate, false positive rate, average case resolution time) shall be performed quarterly and presented to the Board Risk Committee.

6.3 Key Performance Indicators (KPIs)

- False Positive Rate (overall alert population): Target < 12%
- Tier 1 High-Risk Anomaly detection rate (known fraud patterns): Target > 88%
- Average time from alert creation to final disposition (Tier 1 cases): Target < 3.5 hours
- AUTO-DISMISS accuracy (confirmed legitimate on 5% QA sample): Target > 97%
- SAR filing timeliness: 100% within regulatory deadline

6.4 Training and Awareness

All human analysts must complete initial onboarding training on this policy and pass a scenario-based assessment before handling live cases. Annual refresher training is mandatory. AI Agent prompt libraries and decision logic are version-controlled and subject to the same change-management rigor as production code.

APPROVAL

Chief Compliance Officer

Date: _____

Head of Fraud Prevention

Date: _____

APPENDIX A – SAMPLE AI CASE REPORT TEMPLATE

Illustrative Example – Night Velocity Case (Section 4.2)

CASE ID: FRD-2026-0710-884721

Transaction ID: TXN-44981230987

Customer ID: CUST-7729104

Amount / Currency: USD 1,847.00

Merchant: CRYPTOEXCHANGE-INTL (MCC 6051)

Timestamp (Local): 2026-07-10 02:41:17 ET

AI ENRICHMENT NOTES

- 90-day median daily spend: \$312.00 → Current txn = +492% of baseline
- No active Travel Notice on file
- Device fingerprint trust score: 0.31 (new device enrolled 14 hours prior)
- First crypto-related transaction in customer history
- XGBoost probability: 67%
- Isolation Forest anomaly score: 0.89 (top 3% of daily alerts)
- PCA V-14 contribution: +2.8 SD; PCA V-10 contribution: +1.9 SD

CASE SUMMARY

- **Trigger & Quantitative Profile:** Transaction amount \$1,847.00; Isolation Forest anomaly score: 0.89; XGBoost probability: 67%; PCA V-14: +2.8 SD; PCA V-10: +1.9 SD; Local time: 02:41; Velocity vs. 90-day baseline: +492%.
- **Contextual Factors:** No travel notice on file. New device enrollment 14 hours ago with low trust score. First cryptocurrency transaction in 18-month customer history. Merchant MCC 6051 (crypto) classified as high-risk.
- **Initial Risk Assessment:** Strong indicators of potential account takeover or APP fraud involving cryptocurrency conversion. Transaction currently on HOLD.

FINAL RECOMMENDATION: ESCALATE TO HUMAN

APPENDIX B – GLOSSARY OF TERMS

APP Fraud – Authorized Push Payment fraud – victim is tricked into authorizing a payment to a fraudster-controlled account.

ATO – Account Takeover – unauthorized access to and use of a legitimate customer account.

BSA/AML – Bank Secrecy Act / Anti-Money Laundering – U.S. regulatory framework governing financial crime prevention.

CTR – Currency Transaction Report – FinCEN filing required for cash transactions over \$10,000.

FATF – Financial Action Task Force – intergovernmental body setting global AML/CFT standards.

FinCEN – Financial Crimes Enforcement Network – U.S. Treasury bureau responsible for SAR/CTR collection and analysis.

MCC – Merchant Category Code – four-digit classification of merchant business type used in card networks.

PEP – Politically Exposed Person – individual holding prominent public office or close associate thereof.

SAR – Suspicious Activity Report – formal filing with FinCEN when a transaction is suspected to involve illicit activity.

Structuring – Deliberate breaking of large transactions into smaller amounts to evade reporting thresholds (also called smurfing).

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